# Trig Ceva and the ratio lemma

L4-X1 Level 4 Olympiad · Geometry · Beyond the textbook

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Trigonometric Ceva with proof, the ratio lemma and its uniqueness companion, and isogonal lines, for concurrence and for angle problems that angle chasing cannot finish.
**By the end you can** turn angles at a vertex into a ratio on the opposite side, and back, with the ratio lemma; prove concurrence with trigonometric Ceva and find unknown angles with it, including the uniqueness step; recognise isogonal lines and use Steiner's ratio and isogonal conjugates.
**Prerequisites:** Olympiad angle chasing (L4-GE1), Trigonometry and coordinates (L3-GE2). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

## 1 Lesson

*Prove that the three altitudes of an acute triangle $ABC$ are concurrent.* Let the altitudes be $AD$, $BE$, $CF$. In the right-angled triangle $ABD$ we have $\angle BAD = 90^\circ - B$, and in the same way $\angle DAC = 90^\circ - C$, $\angle CBE = 90^\circ - C$, $\angle EBA = 90^\circ - A$, $\angle ACF = 90^\circ - A$ and $\angle FCB = 90^\circ - B$. Hence

$$\frac{\sin\angle BAD}{\sin\angle DAC}\cdot\frac{\sin\angle CBE}{\sin\angle EBA}\cdot\frac{\sin\angle ACF}{\sin\angle FCB} = \frac{\cos B}{\cos C}\cdot\frac{\cos C}{\cos A}\cdot\frac{\cos A}{\cos B} = 1,$$

and by trigonometric Ceva (Theorem 3 below) the altitudes concur. No length on $BC$ was computed.

Ceva's theorem (Giovanni Ceva, 1678) is taught in school with the lengths $BD$, $DC$, $CE$, $EA$, $AF$, $FB$. Olympiad problems usually give angles at the vertices instead, and then these lengths are unknown. The sine form of Ceva, and the lemma behind it, which olympiad trainers call the *ratio lemma*, convert angles at a vertex into a ratio on the opposite side and back. Both are folklore in olympiad training; L4-GE2 states them, and here we prove them, prove the uniqueness statement that most solutions need and most textbooks omit, and use them on isogonal lines. The sine rule and area formula (L3-GE2) and angle chasing (L4-GE1) are assumed.

### 1.1 1. The ratio lemma

**Theorem 1 (ratio lemma).**
Let $D$ be a point of line $BC$ other than $B$ and $C$. Then

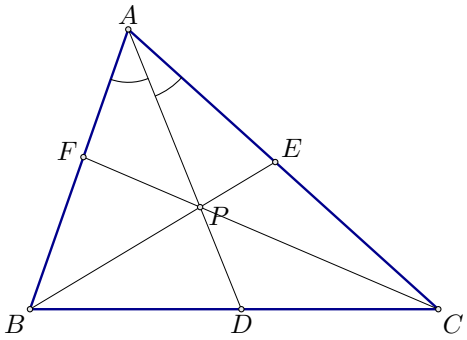
$$\frac{BD}{DC} = \frac{AB\sin\angle BAD}{AC\sin\angle DAC}.$$

*Proof.*  The sine rule in triangles $ABD$ and $ADC$ gives $\dfrac{BD}{\sin\angle BAD} = \dfrac{AB}{\sin\angle ADB}$ and $\dfrac{DC}{\sin\angle DAC} = \dfrac{AC}{\sin\angle ADC}$. The angles $\angle ADB$ and $\angle ADC$ are supplementary if $D$ lies between $B$ and $C$, and equal otherwise; in both cases their sines are equal. Dividing the two equations gives the result. ■



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Two special cases are worth remembering. If $AD$ bisects angle $A$, we get the angle bisector theorem $BD/DC = AB/AC$. If $M$ is the midpoint of $BC$, we get $AB\sin\angle BAM = AC\sin\angle MAC$: a median is a statement about angles at $A$.




**Lemma 2 (uniqueness).**
Let $0^\circ < \theta < 180^\circ$. On the interval $0^\circ < x < \theta$ the function $f(x) = \dfrac{\sin x}{\sin(\theta - x)}$ is increasing. Hence if $f(x) = f(y)$ with $x, y$ in this interval, then $x = y$.

*Proof.*  Expanding $\sin(\theta - x) = \sin\theta\cos x - \cos\theta\sin x$ gives $\dfrac{1}{f(x)} = \sin\theta\cot x - \cos\theta$. Since $\sin\theta > 0$ and $\cot x$ decreases on $(0^\circ, 180^\circ)$, the positive quantity $1/f(x)$ decreases as $x$ increases. ■

Lemma 2 is the step most scripts omit. Trig Ceva gives an equation for an unknown angle; Lemma 2 says the equation has one solution, so a value that satisfies it is the answer.

## 1.2 2. Trigonometric Ceva

**Theorem 3 (trigonometric Ceva).**
Let $D, E, F$ lie inside the sides $BC, CA, AB$ of triangle $ABC$. Then $AD, BE, CF$ are concurrent if and only if

$$\frac{\sin\angle BAD}{\sin\angle DAC}\cdot\frac{\sin\angle CBE}{\sin\angle EBA}\cdot\frac{\sin\angle ACF}{\sin\angle FCB} = 1.$$

*Proof.*  By Theorem 1,

$$\frac{BD}{DC} = \frac{AB\sin\angle BAD}{AC\sin\angle DAC'}, \quad \frac{CE}{EA} = \frac{BC\sin\angle CBE}{BA\sin\angle EBA'}, \quad \frac{AF}{FB} = \frac{CA\sin\angle ACF}{CB\sin\angle FCB}.$$

Multiplying, the side lengths cancel, so the product of sines equals $\frac{BD}{DC}\cdot\frac{CE}{EA}\cdot\frac{AF}{FB}$, and Ceva's theorem gives both directions. ■

For a point $P$ inside the triangle we write the condition as $\dfrac{\sin\angle PAB}{\sin\angle PAC}\cdot\dfrac{\sin\angle PBC}{\sin\angle PBA}\cdot\dfrac{\sin\angle PCA}{\sin\angle PCB} = 1$. Read it by going round the triangle: at each vertex the angle in the numerator is the one next to the side you reach first going $A \to B \to C \to A$.

**Corollary 4 (Ceva on a circle).**
Let $A_1, B_1, C_1$ lie on the circumcircle of $ABC$, on the arcs $BC, CA, AB$ not containing $A, B, C$.



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Then $AA_1$, $BB_1$, $CC_1$ are concurrent if and only if $A_1B \cdot B_1C \cdot C_1A = A_1C \cdot B_1A \cdot C_1B$.

*Proof.*  The arc condition says that $AA_1$ meets side $BC$ at an interior point, and so on. By the extended sine rule $A_1B = 2R\sin\angle BAA_1$ and $A_1C = 2R\sin\angle A_1AC$, so the condition is Theorem 3. ■

**Example 1.**
A point $P$ inside triangle $ABC$ satisfies $\angle PBC = 18^\circ$, $\angle PBA = 30^\circ$, $\angle PCB = 24^\circ$ and $\angle PCA = 54^\circ$. Find $\angle PAB$.

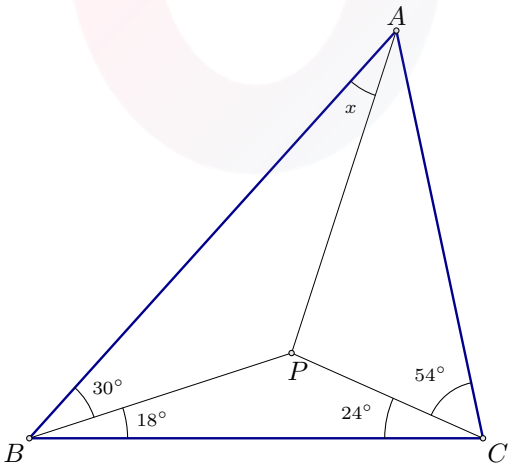
*Solution.*
We have $B = 48^\circ$, $C = 78^\circ$, so $A = 54^\circ$. Let $x = \angle PAB$, so $\angle PAC = 54^\circ - x$. Angle chasing first: $\angle BPC = 138^\circ$, $\angle APB = 150^\circ - x$ and $\angle APC = 72^\circ + x$. These add up to $360^\circ$ for every $x$, so angle chasing produces no equation for $x$. A synthetic solution needs an auxiliary point, and there is no obvious one.
Trig Ceva gives an equation at once:

$$\frac{\sin x}{\sin(54^\circ - x)} = \frac{\sin 30^\circ \sin 24^\circ}{\sin 18^\circ \sin 54^\circ}.$$

A calculator gives about $0.813$ for the right side, and $x = 24^\circ$ fits; we prove it. Since $\sin 54^\circ = \cos 36^\circ$,

$$\sin 18^\circ \cos 36^\circ = \frac{2\sin 18^\circ \cos 18^\circ \cos 36^\circ}{2\cos 18^\circ} = \frac{\sin 36^\circ \cos 36^\circ}{2\cos 18^\circ} = \frac{\sin 72^\circ}{4\cos 18^\circ} = \frac{1}{4}.$$

So the right side is $\frac{1}{2}\sin 24^\circ \cdot 4 = 2\sin 24^\circ = \dfrac{\sin 24^\circ}{\sin 30^\circ}$, which is $f(24^\circ)$ with $\theta = 54^\circ$. By Lemma 2, $x = 24^\circ$ is the only solution: $\angle PAB = 24^\circ$ and $\angle PAC = 30^\circ$.




**Example 2.**
The incircle of triangle $ABC$ touches $BC$, $CA$, $AB$ at $D$, $E$, $F$. Let $X$, $Y$, $Z$ be the feet of the perpendiculars from $D$ to $EF$, from $E$ to $FD$ and from $F$ to $DE$. Prove that $AX$, $BY$, $CZ$ are concurrent.

*Solution.*
Length Ceva would need the points where $AX$ meets $BC$, which are hard to find. The angles

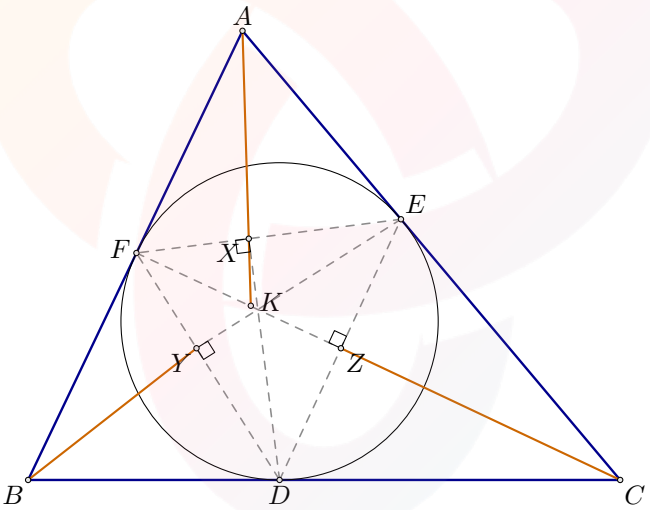


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at $A$ are accessible through triangle $AEF$, in which $AX$ is a cevian, so we use the ratio lemma there.
The angles of triangle $DEF$ are $\angle EDF = 90^\circ - \frac{A}{2}$, $\angle DEF = 90^\circ - \frac{B}{2}$, $\angle DFE = 90^\circ - \frac{C}{2}$ (tangent-chord angle; for instance $\angle DEF = \angle BDF = 90^\circ - \frac{B}{2}$ as $BD = BF$). All are acute, so $X$ lies inside segment $EF$ and $X$ is inside triangle $AEF$. The incircle is the circumcircle of $DEF$, so with $r$ its radius, $DE = 2r\cos\frac{C}{2}$ and $DF = 2r\cos\frac{B}{2}$. Then

$$\frac{EX}{XF} = \frac{DE\cos\angle DEF}{DF\cos\angle DFE} = \frac{\cos\frac{C}{2}\sin\frac{B}{2}}{\cos\frac{B}{2}\sin\frac{C}{2}} = \frac{\tan\frac{B}{2}}{\tan\frac{C}{2}}.$$

In triangle $AEF$ we have $AE = AF$, so the ratio lemma gives $\frac{EX}{XF} = \frac{\sin\angle EAX}{\sin\angle XAF} = \frac{\sin\angle XAC}{\sin\angle BAX}$. Hence $\frac{\sin\angle BAX}{\sin\angle XAC} = \frac{\tan\frac{C}{2}}{\tan\frac{B}{2}}$, and in the same way $\frac{\sin\angle CBY}{\sin\angle YBA} = \frac{\tan\frac{A}{2}}{\tan\frac{C}{2}}$ and $\frac{\sin\angle ACZ}{\sin\angle ZCB} = \frac{\tan\frac{B}{2}}{\tan\frac{A}{2}}$. The product is 1. Each line $AX$, $BY$, $CZ$ meets the opposite side at an interior point, since $X$ is inside the triangle, so Theorem 3 shows that they are concurrent.




## 1.3 3. Isogonal lines

Two lines through $A$ are *isogonal* if each is the reflection of the other in the bisector of angle $A$; for lines meeting segment $BC$ at $D$ and $E$ this says $\angle BAD = \angle EAC$. The reflection of the median from $A$ is the $A$-symmedian. The altitude $AH$ and the line $AO$ are isogonal: $\angle BAH = 90^\circ - B$, and triangle $AOC$ is isosceles with $\angle AOC = 2B$, so $\angle OAC = 90^\circ - B$ (for an acute triangle; directed angles handle the rest).

**Theorem 5 (Steiner).**
If $D$ and $E$ lie on segment $BC$ and $AD$, $AE$ are isogonal, then

$$\frac{BD}{DC}\cdot\frac{BE}{EC} = \frac{AB^2}{AC^2}.$$

**Theorem 6 (isogonal conjugate).** If $P$ is inside triangle $ABC$, the reflections of $AP$, $BP$, $CP$ in the bisectors of the angles at $A$, $B$, $C$ meet at one point $P^*$.



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*Proofs.* For Theorem 5, apply Theorem 1 to $D$ and to $E$ and multiply; since $\angle BAD = \angle EAC$ and $\angle DAC = \angle BAE$, the sines cancel and $AB^2/AC^2$ remains. For Theorem 6, reflecting the three lines swaps the two angles at each vertex, so each fraction in Theorem 3 is replaced by its reciprocal. The product is still 1 and the reflected lines are still inside the angles, so they concur. ■

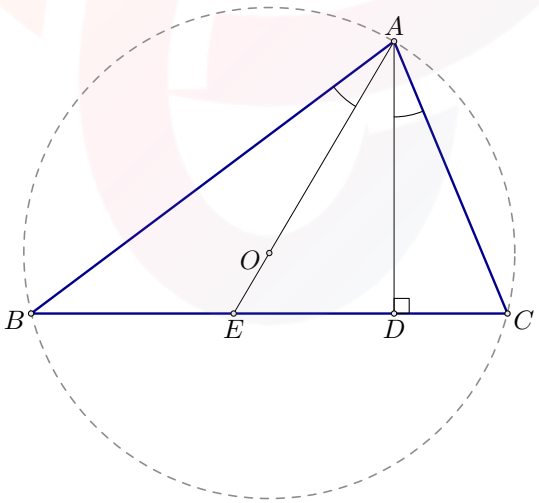
So $O$ and $H$ are isogonal conjugates, the incentre is its own conjugate, and the conjugate of the centroid is the *symmedian point*.

**Example 3.**
Triangle $ABC$ has $AB = 20$, $AC = 13$, $BC = 21$, and circumcentre $O$. Line $AO$ meets $BC$ at $E$. Find $BE$.

*Solution.*
The standard way is coordinates. With $B = (0,0)$, $C = (21,0)$ we find $A = (16,12)$, and $O = (\frac{21}{2}, y)$ with $OB = OA$, which gives $y = \frac{8}{3}$. Line $AO$ reaches $y = 0$ at $x = 16 - \frac{9}{7} \cdot \frac{11}{2} = \frac{125}{14}$.
With isogonal lines we never find $O$. The foot $D$ of the altitude from $A$ has $BD = 16$ and $DC = 5$ (from the coordinates, or from $AB^2 - BD^2 = AC^2 - DC^2$). Since $AD$ and $AE$ are isogonal and the triangle is acute (as $21^2 < 20^2 + 13^2$), Theorem 5 gives

$$\frac{BE}{EC} = \frac{20^2}{13^2} \cdot \frac{DC}{BD} = \frac{400}{169} \cdot \frac{5}{16} = \frac{125}{169},$$

so $BE = 21 \cdot \frac{125}{294} = \frac{125}{14}$. The second method works equally well when the triangle has no convenient coordinates.



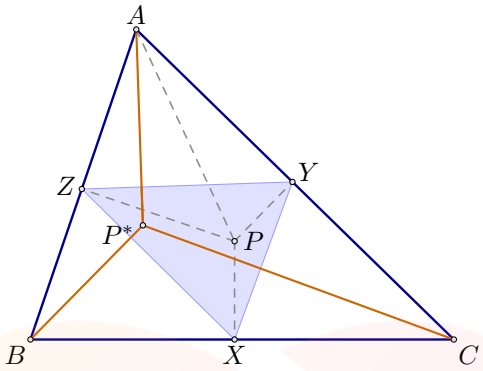

**Example 4.**
Let $P$ be a point inside an acute triangle $ABC$, and let $X, Y, Z$ be the feet of the perpendiculars from $P$ to $BC$, $CA$, $AB$. Prove that the line through $A$ perpendicular to $YZ$, the line through $B$ perpendicular to $ZX$ and the line through $C$ perpendicular to $XY$ are concurrent.

*Solution.*
Three perpendiculars to the sides of a triangle other than $ABC$ do not give a usable Ceva configuration directly; we identify the lines instead. Since $\angle AZP = \angle AYP = 90°$, the points $A$, $Z$, $P$, $Y$ lie on the circle with diameter $AP$. As the triangle is acute, the foot $Z$ lies inside segment $AB$, so $Z$ and $P$ both lie on the same side of line $AC$ as $B$, and $\angle AZY = \angle APY =$



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$90^\circ - \angle PAY$, which is acute.
Let $\ell_A$ be the line through $A$ perpendicular to $YZ$, meeting it at $U$. In the right-angled triangle $AZU$, the angle between $\ell_A$ and $AB$ is $90^\circ - \angle AZY = \angle PAY = \angle PAC$. So $\ell_A$ is the reflection of $AP$ in the bisector of angle $A$. In the same way the other two lines are the reflections of $BP$ and $CP$, and by Theorem 6 they meet at the isogonal conjugate $P^*$.




**Common traps.**

- Pairing the angles wrongly in Theorem 3. Write the six angles in cyclic order before substituting.
- Stating the angle found by trig Ceva without Lemma 2. A marker wants to see why there is only one solution.
- Using the unsigned forms for points outside the triangle. If $A_1$ lies on the same arc $BC$ as $A$ with $AA_1 \parallel BC$, and similarly for $B_1$, $C_1$, then $A_1B = AC$, $A_1C = AB$ and so on, the product condition of Corollary 4 holds, and yet the three lines through the vertices parallel to the opposite sides are not concurrent.
- In Theorem 5, check whether $D$ and $E$ lie on the segment or on its extension; Theorem 1 holds in both cases, but the angle equalities need checking.

### 1.4 How to spot this in a contest

1. A point inside a triangle given by angles at two vertices, with the angle at the third vertex asked for: trig Ceva, then Lemma 2.
2. An angle chase in which the angle sum at a point is automatically $360^\circ$ and gives nothing.
3. Concurrency of cevians whose feet on the opposite sides are hard to locate, for example because they are defined inside another triangle or on the circumcircle.
4. A ratio $BD/DC$ wanted with the angles at $A$ known, or the reverse.
5. Reflections in angle bisectors, or the pairs altitude and circumcentre, median and tangents: isogonal lines and Theorems 5 and 6.

## 2 Practice problems



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The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

**Problem 1** ⋆ Triangle $ABC$ has $\angle BAC = 90°$, $AB = 5$ and $AC = 12$. The point $D$ on $BC$ satisfies $\angle BAD = 30°$. Find $BD/DC$.

**Problem 2** ⋆⋆ A point $P$ inside triangle $ABC$ satisfies $\angle PBC = 24°$, $\angle PBA = 30°$, $\angle PCB = 36°$ and $\angle PCA = 30°$. Find $\angle PAB$.

**Problem 3** ⋆⋆ Triangle $ABC$ has $AB = 6$, $AC = 4$ and $BC = 5$. The median from $A$ meets $BC$ at $M$, and the $A$-symmedian (the reflection of line $AM$ in the bisector of angle $A$) meets $BC$ at $K$. Find $MK$.

**Problem 4** ⋆⋆⋆ Let $0° < \theta < 30°$. Triangle $ABC$ has $\angle ABC = 3\theta$ and $\angle ACB = 120° - \theta$. The point $P$ inside the triangle satisfies $\angle PBC = \theta$ and $\angle PCB = 30°$. Prove that $AP$ bisects angle $BAC$.

**Problem 5** ⋆⋆⋆ Triangle $ABC$ has all three angles between 45° and 90°. The altitudes are $AD$, $BE$, $CF$, and $X$, $Y$, $Z$ are the feet of the perpendiculars from $D$ to $EF$, from $E$ to $FD$ and from $F$ to $DE$. Prove that $AX$, $BY$, $CZ$ are concurrent.

**Problem 6** ⋆⋆⋆⋆ Let $ABC$ be a triangle and let $\alpha, \beta, \gamma$ be positive angles with $A + \alpha$, $B + \beta$, $C + \gamma$ all less than 180°, and $\beta + \gamma$, $\gamma + \alpha$, $\alpha + \beta$ all less than 180°. Points $X$, $Y$, $Z$ lie on the opposite sides of $BC$, $CA$, $AB$ from $A$, $B$, $C$ respectively, with

$$\angle XBC = \angle ZBA = \beta, \qquad \angle XCB = \angle YCA = \gamma, \qquad \angle YAC = \angle ZAB = \alpha.$$

Prove that $AX$, $BY$, $CZ$ are concurrent.

## 3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AHSME 1998 Problem 28** ⋆⋆
   *Ratio lemma at A with the angles theta and 2 theta.*
2. **AMC 12 2001 Problem 24** ⋆⋆
   *Call the unknown angle x and apply the sine rule in triangles ABD and ADC.*
3. **AIME I 2012 Problem 12** ⋆⋆
   *Use the ratio lemma with the 30 degree angles at C, once for D and once for E.*
4. **BMO2 2004/05 Problem 2** ⋆⋆
   *Show that DE bisects angle ADC by comparing AE/EC with AD/DC.*
5. **AIME 1999 Problem 14** ⋆⋆⋆
   *Trig Ceva with three equal angles, or split the area at P.*
6. **AIME II 2005 Problem 14** ⋆⋆⋆
   *AD and AE are isogonal: Theorem 5.*



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7. **BMO1 2024/25 Problem 4** ⋆⋆⋆
   *P lies on the median, so the ratio lemma gives a relation between the angles at A.*

8. **USAMO 1996 Problem 5** ⋆⋆⋆
   *Trig Ceva for the unknown angle at B, then Lemma 2.*

9. **AIME II 2010 Problem 15** ⋆⋆⋆⋆
   *Find the ratio of the sines of the angles BAQ and QAC from triangles at P, then apply the ratio lemma.*

10. **BMO2 2016/17 Problem 3** ⋆⋆⋆⋆
    *Angle bisector theorem twice; compare the two ratios with the sine rule.*

11. **USAMO 2008 Problem 2** ⋆⋆⋆⋆
    *Compute the angles ABF and ACF, then identify line AF with trig Ceva or Theorem 5.*



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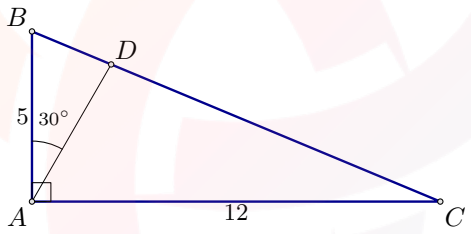
## 4 Hints

1. Theorem 1 with the angles at $A$.
2. Write down trig Ceva with $x = \angle PAB$ and look for cancellation. Then use Lemma 2.
3. Theorem 5 with $D = M$.
4. Find all six angles at $B$ and $C$, then simplify $\sin\theta\cos\theta$.
5. Copy Example 2, with the orthic triangle in place of the contact triangle. You need the angles and the sides of triangle $DEF$.
6. Let $AX$ meet $BC$ at $D$ and compute $BD/DC$ as a ratio of the areas of $ABX$ and $ACX$.

## 5 Solutions

### Problem 1

*Triangle $ABC$ has $\angle BAC = 90°$, $AB = 5$ and $AC = 12$. The point $D$ on $BC$ satisfies $\angle BAD = 30°$. Find $BD/DC$.*




Here $\angle DAC = 90° - 30° = 60°$. By the ratio lemma,

$$\frac{BD}{DC} = \frac{AB\sin 30°}{AC\sin 60°} = \frac{5\cdot\frac{1}{2}}{12\cdot\frac{\sqrt{3}}{2}} = \frac{5}{12\sqrt{3}} = \frac{5\sqrt{3}}{36}.$$

The hypotenuse $BC = 13$ was not needed.

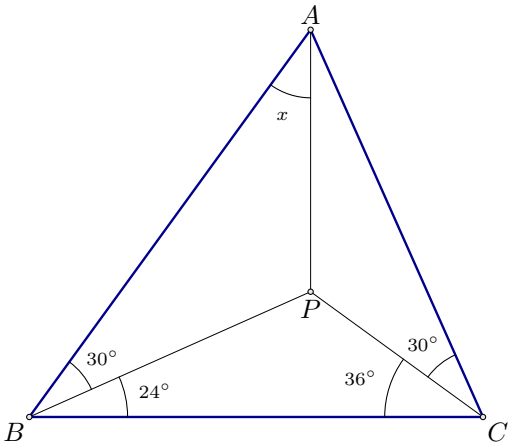
**Answer:** $\dfrac{5\sqrt{3}}{36}$

### Problem 2

*A point $P$ inside triangle $ABC$ satisfies $\angle PBC = 24°$, $\angle PBA = 30°$, $\angle PCB = 36°$ and $\angle PCA = 30°$. Find $\angle PAB$.*



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We have $B = 54^\circ$, $C = 66^\circ$ and $A = 60^\circ$. Let $x = \angle PAB$, so $\angle PAC = 60^\circ - x$. Trig Ceva gives

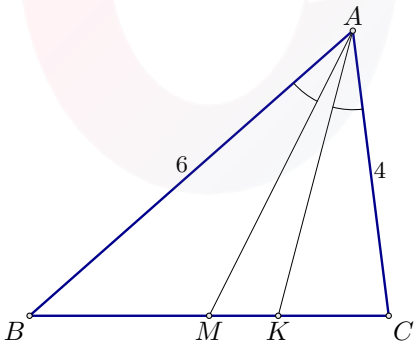
$$\frac{\sin x}{\sin(60^\circ - x)} \cdot \frac{\sin 24^\circ}{\sin 30^\circ} \cdot \frac{\sin 30^\circ}{\sin 36^\circ} = 1, \quad \text{so} \quad \frac{\sin x}{\sin(60^\circ - x)} = \frac{\sin 36^\circ}{\sin 24^\circ}.$$

The value $x = 36^\circ$ satisfies this, since then $60^\circ - x = 24^\circ$. By Lemma 2 with $\theta = 60^\circ$ it is the only solution in $(0^\circ, 60^\circ)$, so $\angle PAB = 36^\circ$.

**Answer:** $36^\circ$

### Problem 3

*Triangle $ABC$ has $AB = 6$, $AC = 4$ and $BC = 5$. The median from $A$ meets $BC$ at $M$, and the $A$-symmedian (the reflection of line $AM$ in the bisector of angle $A$) meets $BC$ at $K$. Find $MK$.*




The symmedian meets segment $BC$, since reflection in the bisector keeps a line inside angle $A$. By Theorem 5 with $BM/MC = 1$,

$$\frac{BK}{KC} = \frac{AB^2}{AC^2} = \frac{36}{16} = \frac{9}{4},$$

so $BK = 5 \cdot \frac{9}{13} = \frac{45}{13}$. As $BM = \frac{5}{2}$ and $\frac{45}{13} > \frac{5}{2}$, the point $K$ lies between $M$ and $C$, and $MK = \frac{45}{13} - \frac{5}{2} = \frac{90-65}{26} = \frac{25}{26}$.

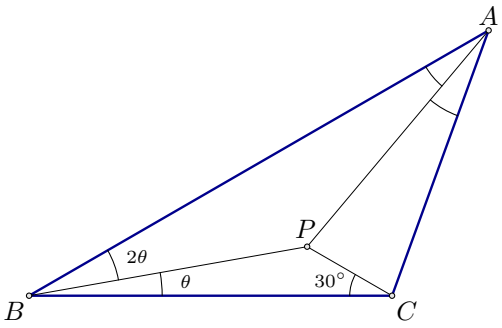
**Answer:** $\dfrac{25}{26}$



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### Problem 4

*Let $0^\circ < \theta < 30^\circ$. Triangle $ABC$ has $\angle ABC = 3\theta$ and $\angle ACB = 120^\circ - \theta$. The point $P$ inside the triangle satisfies $\angle PBC = \theta$ and $\angle PCB = 30^\circ$. Prove that $AP$ bisects angle $BAC$.*




The angles of the triangle are $B = 3\theta$, $C = 120^\circ - \theta$ and $A = 60^\circ - 2\theta$, all positive for $0^\circ < \theta < 30^\circ$. The point $P$ exists and is inside the triangle, since $\theta < B$ and $30^\circ < C$. We have $\angle PBA = 3\theta - \theta = 2\theta$ and $\angle PCA = 120^\circ - \theta - 30^\circ = 90^\circ - \theta$. Let $x = \angle PAB$, so $\angle PAC = A - x$ with $0^\circ < x < A$.

By Theorem 3,

$$\frac{\sin x}{\sin(A - x)} \cdot \frac{\sin\theta}{\sin 2\theta} \cdot \frac{\sin(90^\circ - \theta)}{\sin 30^\circ} = 1.$$

Now $\sin\theta \sin(90^\circ - \theta) = \sin\theta\cos\theta = \frac{1}{2}\sin 2\theta = \sin 30^\circ \sin 2\theta$, so the product of the last two fractions is 1. Hence $\dfrac{\sin x}{\sin(A - x)} = 1 = \dfrac{\sin\frac{A}{2}}{\sin\frac{A}{2}}$, and by Lemma 2 (with $\theta$ there equal to $A$) we get $x = \frac{A}{2}$. So $AP$ bisects angle $BAC$.

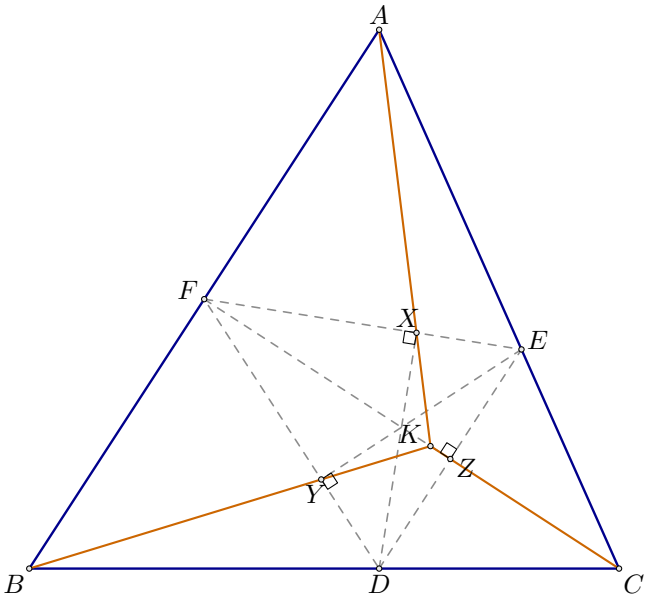
The angle chase alone gives $\angle BPC = 150^\circ - \theta$ and an equation in $x$ that holds for every $x$, so some trigonometric or auxiliary-point argument is needed.

### Problem 5

*Triangle $ABC$ has all three angles between $45^\circ$ and $90^\circ$. The altitudes are $AD$, $BE$, $CF$, and $X$, $Y$, $Z$ are the feet of the perpendiculars from $D$ to $EF$, from $E$ to $FD$ and from $F$ to $DE$. Prove that $AX$, $BY$, $CZ$ are concurrent.*



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We use the standard facts about the orthic triangle of an acute triangle: triangles $AEF$, $DBF$, $DEC$ are similar to $ABC$, with $AE = AB\cos A$ and $AF = AC\cos A$; the sides of $DEF$ are $EF = a\cos A$, $FD = b\cos B$, $DE = c\cos C$; and its angles are $\angle EDF = 180^\circ - 2A$, $\angle DEF = 180^\circ - 2B$, $\angle DFE = 180^\circ - 2C$. (For instance $\angle AEF = B$ and $\angle CED = B$, so $\angle DEF = 180^\circ - 2B$.)

All angles of $ABC$ exceed $45^\circ$, so all angles of $DEF$ are acute, and $X$ lies inside segment $EF$, hence inside triangle $AEF$. Then

$$\frac{EX}{XF} = \frac{DE\cos\angle DEF}{DF\cos\angle DFE} = \frac{c\cos C\cdot(-\cos 2B)}{b\cos B\cdot(-\cos 2C)} = \frac{c\cos C\cos 2B}{b\cos B\cos 2C}.$$

By the ratio lemma in triangle $AEF$,

$$\frac{EX}{XF} = \frac{AE\sin\angle EAX}{AF\sin\angle XAF} = \frac{c\,\sin\angle XAC}{b\,\sin\angle BAX}.$$

Comparing, $\dfrac{\sin\angle BAX}{\sin\angle XAC} = \dfrac{\cos B\cos 2C}{\cos C\cos 2B}$. In the same way $\dfrac{\sin\angle CBY}{\sin\angle YBA} = \dfrac{\cos C\cos 2A}{\cos A\cos 2C}$ and $\dfrac{\sin\angle ACZ}{\sin\angle ZCB} = \dfrac{\cos A\cos 2B}{\cos B\cos 2A}$.

The product of the three fractions is 1. Each of $AX$, $BY$, $CZ$ meets the opposite side at an interior point, since $X$, $Y$, $Z$ are inside the triangle, so by Theorem 3 the three lines are concurrent.

The hypothesis on the angles is used once, to put $X$, $Y$, $Z$ inside the sides of $DEF$. Without it one of the feet lies outside, and a signed version of the argument is needed.

### Problem 6

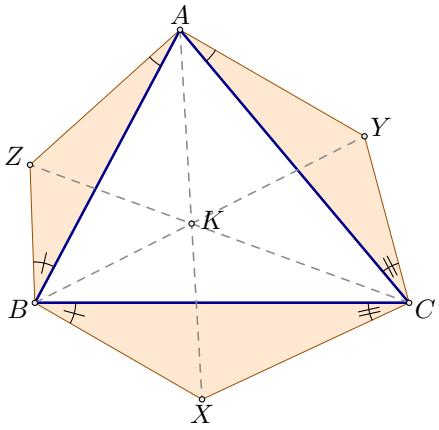
*Let $ABC$ be a triangle and let $\alpha$, $\beta$, $\gamma$ be positive angles with $A+\alpha$, $B+\beta$, $C+\gamma$ all less than $180^\circ$, and $\beta+\gamma$, $\gamma+\alpha$, $\alpha+\beta$ all less than $180^\circ$. Points $X$, $Y$, $Z$ lie on the opposite sides of $BC$, $CA$, $AB$ from $A$, $B$, $C$ respectively, with*

$$\angle XBC = \angle ZBA = \beta, \qquad \angle XCB = \angle YCA = \gamma, \qquad \angle YAC = \angle ZAB = \alpha.$$

*Prove that $AX$, $BY$, $CZ$ are concurrent.*



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The conditions $\beta+\gamma<180^\circ$ etc. make the triangles $XBC$, $YCA$, $ZAB$ exist. Since $X$ is on the other side of $BC$ from $A$, and $\angle ABX = B+\beta$ and $\angle ACX = C+\gamma$ are less than $180^\circ$, the quadrilateral $ABXC$ is convex, so its diagonals $AX$ and $BC$ meet at a point $D$ inside both. Similarly $BY$ meets $CA$ at an interior point $E$ and $CZ$ meets $AB$ at an interior point $F$.

The distances from $B$ and $C$ to line $AX$ are in the ratio $BD:DC$, so

$$\frac{BD}{DC}=\frac{[ABX]}{[ACX]}=\frac{\frac12 AB\cdot BX\sin(B+\beta)}{\frac12 AC\cdot CX\sin(C+\gamma)}.$$

The sine rule in triangle $XBC$ gives $BX/CX=\sin\gamma/\sin\beta$. Hence, writing $a,b,c$ for the sides,

$$\frac{BD}{DC}=\frac{c\sin\gamma\sin(B+\beta)}{b\sin\beta\sin(C+\gamma)}.$$

The same computation at the other vertices gives

$$\frac{CE}{EA}=\frac{a\sin\alpha\sin(C+\gamma)}{c\sin\gamma\sin(A+\alpha)},\qquad \frac{AF}{FB}=\frac{b\sin\beta\sin(A+\alpha)}{a\sin\alpha\sin(B+\beta)}.$$

Every factor cancels in the product, so $\frac{BD}{DC}\cdot\frac{CE}{EA}\cdot\frac{AF}{FB}=1$, and since $D$, $E$, $F$ are interior points of the sides, Ceva's theorem shows that $AX$, $BY$, $CZ$ are concurrent. ■

This is Jacobi's theorem. With $\alpha=\beta=\gamma$ the triangles $XBC$, $YCA$, $ZAB$ are isosceles and similar (Kiepert's theorem); the value $30^\circ$ gives the lines from the vertices to the centres of the outer equilateral triangles.